

A Lacunary Martingale-Model Endpoint for the Subordinated Poisson Square Function

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2026-07-18

Source Question

The source paper is Zhendong Xu and Hao Zhang, *From the Littlewood-Paley-Stein Inequality to the Burkholder-Gundy Inequality*, arXiv:2111.05164. Its introduction records that Xu had proved

$$\mathbb{L}_p^{\mathbb{P}} \lesssim \max\{p, \sqrt{p'}\}$$

for the reverse Littlewood-Paley-Stein inequality of the Poisson semigroup subordinated to a symmetric diffusion semigroup, and that the corresponding problem for $\mathbb{L}_p^{\mathbb{P}}$ remains open at the endpoint $p \rightarrow 1$. Equivalently, Xu's Problem 8.4 in arXiv:2105.12175 asks, in particular, whether for a symmetric submarkovian or Markovian semigroup there is a semigroup-dependent constant C such that

$$\|f - Ff\|_{L^1(\Omega)} \leq C \|G_2^{\mathbb{P}}(f)\|_{L^1(\Omega)}.$$

Result Type

This packet is a partial result. It proves the endpoint estimate for a martingale-generated model class, with sufficiently lacunary generator spectrum. It does not settle the arbitrary symmetric submarkovian case.

Definitions

Let $(\Omega, \mathcal{A}, \mu)$ be a probability space equipped with an increasing filtration $(\mathcal{A}_n)_{n \geq 1}$. Let E_n denote conditional expectation onto $\mathcal{A}_n$, set $D_n = E_n - E_{n-1}$ for $n \geq 2$, and write $F = E_1$. For a martingale terminal value f , define

$$S(f - Ff) = \left(\sum_{n \geq 2} |D_n f|^2 \right)^{1/2}.$$

Let $0 = \lambda_1 < \lambda_2 < \lambda_3 < \dots$, $\lambda_n \rightarrow \infty$, and define on martingale differences

$$\mathbb{T}_t f = Ff + \sum_{n \geq 2} e^{-\lambda_n t} D_n f, \quad t > 0.$$

This is the Xu-Zhang martingale construction from arXiv:2111.05164, written in generator form: choosing $a_{n-1} = 1 - e^{-\lambda_n}$ gives their $\mathbb{T}_t = T^t$. In particular $(\mathbb{T}_t)_{t > 0}$ is a symmetric diffusion semigroup. Its subordinated Poisson semigroup is

$$\mathbb{P}_t f = Ff + \sum_{n \geq 2} e^{-t\sqrt{\lambda_n}} D_n f.$$

The Poisson g -function is

$$G^P(f) = \left(\int_0^\infty \left| t \frac{\partial}{\partial t} P_t f \right|^2 \frac{dt}{t} \right)^{1/2}.$$

Proof Intuition

On each martingale difference $D_n f$, the Poisson derivative contributes the scalar kernel $t\sqrt{\lambda_n}e^{-t\sqrt{\lambda_n}}$. If the numbers $\sqrt{\lambda_n}$ are sufficiently separated, these kernels form a Riesz sequence in $L^2((0, \infty), dt/t)$. Therefore the Poisson square function is pointwise equivalent to the usual martingale square function. Davis' endpoint square-function inequality then gives the desired L^1 reverse estimate.

Lemma 1 (Lacunary Poisson kernels). *Let $\alpha_{n+1}/\alpha_n \geq \rho > 9$, and define*

$$h_n(t) = 2t\alpha_n e^{-t\alpha_n}, \quad t > 0,$$

as elements of $H = L^2((0, \infty), dt/t)$. Put $\eta = 8/(\rho - 1) < 1$. Then for every finitely supported scalar sequence (c_n) ,

$$(1 - \eta) \sum_n |c_n|^2 \leq \left\| \sum_n c_n h_n \right\|_H^2 \leq (1 + \eta) \sum_n |c_n|^2.$$

Proof. The diagonal entries are normalized:

$$\|h_n\|_H^2 = 4\alpha_n^2 \int_0^\infty t e^{-2\alpha_n t} dt = 1.$$

For $m < n$,

$$\langle h_m, h_n \rangle_H = 4\alpha_m \alpha_n \int_0^\infty t e^{-(\alpha_m + \alpha_n)t} dt = \frac{4\alpha_m \alpha_n}{(\alpha_m + \alpha_n)^2}.$$

Since $\alpha_n/\alpha_m \geq \rho^{n-m}$, this is at most $4\rho^{-(n-m)}$. Hence the sum of absolute off-diagonal entries in every row of the Gram matrix is at most

$$2 \sum_{k \geq 1} 4\rho^{-k} = \frac{8}{\rho - 1} = \eta.$$

The conclusion follows from the elementary Gershgorin/Schur bound for this Hermitian Gram matrix. $\square$

Theorem 1 (Endpoint for lacunary martingale Poisson models). *Assume $\sqrt{\lambda_{n+1}/\lambda_n} \geq \rho > 9$ for all $n \geq 2$. Then, pointwise for all finite martingales,*

$$\frac{\sqrt{1-\eta}}{2} S(f - Ff) \leq G^P(f) \leq \frac{\sqrt{1+\eta}}{2} S(f - Ff), \quad \eta = \frac{8}{\rho - 1}.$$

Consequently there is a constant C_ρ , independent of the filtration and of f , such that

$$\|f - Ff\|_{L^1(\Omega)} \leq C_\rho \|G^P(f)\|_{L^1(\Omega)}.$$

In particular, in this model class the reverse Poisson constants remain bounded at the endpoint $p \downarrow 1$.

Proof. For a fixed $\omega \in \Omega$, put $c_n = D_n f(\omega)$ and $\alpha_n = \sqrt{\lambda_n}$. Since

$$t \frac{\partial}{\partial t} P_t f(\omega) = - \sum_{n \geq 2} t \alpha_n e^{-t \alpha_n} D_n f(\omega),$$

we have

$$G^P(f)(\omega) = \frac{1}{2} \left\| \sum_{n \geq 2} c_n h_n \right\|_{L^2(dt/t)}.$$

Lemma 1 gives the asserted pointwise comparison with $(\sum |D_n f(\omega)|^2)^{1/2}$.

Davis' endpoint martingale square-function inequality gives a universal constant C_D such that

$$\|f - Ff\|_1 \leq C_D \|S(f - Ff)\|_1$$

for every scalar martingale. Combining this with the lower pointwise bound for G^P yields

$$\|f - Ff\|_1 \leq \frac{2C_D}{\sqrt{1-\eta}} \|G^P(f)\|_1.$$

The finite-martingale proof extends to the usual L^1 domain by truncating the martingale and applying monotone convergence to the square functions. The bounded $p \downarrow 1$ statement follows from the same Davis-Burkholder-Gundy inequalities, whose reverse square-function constants are bounded on $1 \leq p \leq 2$. $\square$

Verification Notes

The only nontrivial semigroup input is the Xu-Zhang construction: increasing λ_n corresponds to an increasing sequence $a_{n-1} = 1 - e^{-\lambda_n}$, so their Theorem 1.1 gives a symmetric diffusion semigroup with eigenvalue $e^{-\lambda_n t}$ on $D_n L^p$. The Poisson subordination then replaces λ_n by $\sqrt{\lambda_n}$, exactly as used above.

The lacunarity threshold $\rho > 9$ is not intended to be sharp. It is chosen only to make the Riesz-sequence estimate a one-line Gershgorin argument. A sharper Toeplitz or total-positivity analysis may cover much denser spectra, including the numerical regime of Xu-Zhang's displayed 16^n heat rates, but that strengthening is not needed for this packet.

Limitations

This packet does not prove Xu's full Problem 8.4. The proof uses the martingale difference spectral representation and a lacunarity hypothesis on the generator. Arbitrary symmetric diffusion semigroups need not have such a discrete martingale-difference model, and the general semigroup-BMO dual route remains open in this work.

Novelty and Literature Check

Local searches of the run registry, solution index, attempt index, and proof gap index for arXiv:2111.05164, Littlewood-Paley-Stein, Poisson reverse constants, martingale cotype, and symmetric diffusion semigroups found no prior packet for this target. Local source checks included arXiv:2111.05164, arXiv:2105.12175, and arXiv:2105.12795. Web searches on 2026-07-18 for phrases including " L_p^P ",

“ $p \rightarrow 1$ ”, “reverse Littlewood-Paley-Stein”, “Poisson semigroup”, and “semigroup BMO” found Xu’s explicit Problem 8.4 and classical Euclidean positive results, but no recorded theorem proving the lacunary martingale-model endpoint above.

References

- [1] Z. Xu and H. Zhang, *From the Littlewood-Paley-Stein Inequality to the Burkholder-Gundy Inequality*, arXiv:2111.05164; Trans. Amer. Math. Soc. 376 (2023), 371–389.
- [2] Q. Xu, *Holomorphic functional calculus and vector-valued Littlewood-Paley-Stein theory for semigroups*, arXiv:2105.12175.
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- [4] B. Davis, *On the integrability of the martingale square function*, Israel J. Math. 8 (1970), 187–190.
- [5] D. L. Burkholder, *Sharp inequalities for martingales and stochastic integrals*, Asterisque 157–158 (1988), 75–94.

Human Review Recommendation

The packet is recommended for review as a partial positive endpoint theorem. The main checks are: (i) confirm the indexing between a_n , D_n , and λ_n in the Xu-Zhang semigroup; (ii) verify the Gram estimate in Lemma 1; and (iii) confirm the endpoint Davis inequality is being used in the correct scalar martingale form.